Table 1: Every pre-registered decision rule (node.yaml, written before the runs), applied mechanically (analysis/evidence.py). PASS/FAIL: the rule's condition holds / fails and cannot change with the pending jobs; *: provisional (decided on the finished jobs, but further seeds of counted systems are still running and can move the CIs); INCONCL.: pending systems decide; PENDING: no data; REPORT.: descriptive rule (reported, no pass/fail). Outcomes are the single-method re-reading (author decision 2026-10-26: HyperKKL := lora, $r = 4$, no best-of-variants selection); where that changes the pre-registered reading, the original outcome is shown below it (orig.; '(same)' = unchanged). Vote counts k/n are not corrected for multiplicity (6 systems, several tiers and nodes).

node	question	rule (verbatim, abridged)	outcome	evidence (abridged)
dev systems (design decisions)				
R0	Does the v2 input protocol (one amplitude box, per-plant t_ref, id / shape_ood / sequence_ood tiers) train and evaluate cleanly, with no mass...	Pass if no job fails, rejection < 5% per tier, and every conditioned variant beats autonomous on id/clean on every dev system.	**PASS**	no failed jobs: True; max rejection 0.0% (< 5%: True); every conditioned variant beats autonomous on id: 6/6 systems
C1-where	With identical frozen base maps, context encoder (SFM) and a matched 140k-parameter conditioner, which location generalises best to unseen input...	A location is preferred if it beats every other location with a paired 95% CI excluding 0 on the all-families column on $\geq$ 4/6 dev systems, on shape_ood and...	**FAIL** (same)	location rule over latent / concat / film / HyperKKL (dense-subspace ablation excluded): preferred NONE. HyperKKL (r=4) beats latent and concat: shape_ood 5/6; sequence_ood 4/6
C1-ctx	Does the C1 ranking survive when the context encoder's parameters count toward the budget?	Ranking preserved if the C1-where winner stays within CI of the best here on $\geq$ 4/6 systems.	**PASS** (same)	C1-where winner best-or-within-CI: shape_ood 6/6; sequence_ood 6/6 (need 4/6); TOST(±10%) version: shape_ood 6/6; sequence_ood 6/6
C1-budget-s	Is the C1 ranking an artefact of one parameter budget?	Stable if the C1-where winner is best-or-within-CI at both budgets on $\geq$ 4/6 systems.	**PASS** (same)	C1-where winner best-or-within-CI: shape_ood 6/6; sequence_ood 6/6 (need 4/6); TOST(±10%) version: shape_ood 6/6; sequence_ood 6/6
C1-budget-l	Is the C1 ranking an artefact of one parameter budget?	Stable if the C1-where winner is best-or-within-CI at both budgets on $\geq$ 4/6 systems.	**PASS** (same)	C1-where winner best-or-within-CI: shape_ood 6/6; sequence_ood 6/6 (need 4/6); TOST(±10%) version: shape_ood 6/6; sequence_ood 6/6
C1-budget	Is the C1 ranking an artefact of one parameter budget?	Stable if the C1-where winner is best-or-within-CI at both budgets on $\geq$ 4/6 systems.	**PASS** (same)	stable at BOTH budgets: shape_ood 6/6; sequence_ood 6/6
C1-rank1	Is the LoRA ~ hypernetwork tie an artefact of the low-rank choice?	Stable if lora at this rank stays within an equivalence margin of ±10% of lora rank 4 (C1-where) or better, on $\geq$ 4/6 dev systems (sequence_ood).	**FAIL**	lora at this rank within ±10% of rank 4 or better (TOST non-inferiority, 90% CI): 1/6 (need 4/6)
C1-rank16	Is the LoRA ~ hypernetwork tie an artefact of the low-rank choice?	Stable if lora at this rank stays within an equivalence margin of ±10% of lora rank 4 (C1-where) or better, on $\geq$ 4/6 dev systems (sequence_ood).	**FAIL**	lora at this rank within ±10% of rank 4 or better (TOST non-inferiority, 90% CI): 3/6 (need 4/6)
C1-tune-select	With an equal tuning budget per variant (8 configurations: 4 learning rates x 1 architecture knob, 140k parameters, 3000 steps), which...	REPORTED: per variant the argmin of the mean (over the 6 dev systems, seed 0) validation NRMSE (train.<method>.val_nrmse) of its 8 configurations, ties (< 0.5%) to...	PENDING	[PARTIAL DATA: 54/60 jobs done, 6 running] selection.json not written yet (analysis/select_tuned.py)
C1-tuned	With an equal tuning budget per variant, does weight-space conditioning still beat latent injection and the additive residual?	Tuned HyperKKL (the low-rank hypernetwork, 'lora' code path) beats tuned latent AND tuned concat with a paired 95% CI on $\geq$ 4/6 dev and $\geq$ 4/6 test systems, shape_ood...	PENDING	tuned HyperKKL (lora) beats tuned latent and concat: dev pending; test pending (need 4/6 on each of the four)
C1-basewidth-64	Does a stronger or weaker base decoder change where the input should enter?	Report the C1 ranking per width; 'base-dependent' if the winner changes on $\geq$ 3/6.	**PASS** (same)	winner changes on shape_ood 0/6; sequence_ood 0/6 ('base-dependent' if $\geq$ 3/6; PASS = not base-dependent)

Table 1: (continued)

node	question	rule (verbatim, abridged)	outcome	evidence (abridged)
C1-basewidth-256	Does a stronger or weaker base decoder change where the input should enter?	Report the C1 ranking per width; 'base-dependent' if the winner changes on $\geq$ 3/6.	**PASS** (same)	winner changes on shape_ood 0/6; sequence_ood 0/6 ('base-dependent' if $\geq$ 3/6; PASS = not base-dependent)
C1-modelfree	How much of the accuracy needs the model in phase 1?	Model needed if C1-where beats this node with CI excluding 0 on $\geq$ 4/6.	**FAIL** (same)	C1-where beats model-free phase 1 (same method, CI): shape 0/6; seq 0/6 (need 4/6)
C2-floor-025	Corollary 1: input history older than ~ln(1/eta)/alpha barely affects the map. If the context is forced to forget at rate $\geq$ c alpha (Lemma 2 then...	Memory beyond the theory window is unnecessary if C2-floor-05 is within CI of C1-where (no floor) for lora and hyper on $\geq$ 5/6 dev systems, on id and sequence_ood....	**FAIL** (same)	no loss vs no floor (HyperKKL, id & seq): 6/6 (need 5/6); TOST(10%) non-inferior: 3/6; probe decays at $\geq$ floor rate: 1/6 systems
C2-floor-05	Corollary 1: input history older than ~ln(1/eta)/alpha barely affects the map. If the context is forced to forget at rate $\geq$ c alpha (Lemma 2 then...	Memory beyond the theory window is unnecessary if C2-floor-05 is within CI of C1-where (no floor) for lora and hyper on $\geq$ 5/6 dev systems, on id and sequence_ood....	**FAIL** (same)	no loss vs no floor (HyperKKL, id & seq): 5/6 (need 5/6); TOST(10%) non-inferior: 2/6; probe decays at $\geq$ floor rate: 0/6 systems
C2-floor-1	Corollary 1: input history older than ~ln(1/eta)/alpha barely affects the map. If the context is forced to forget at rate $\geq$ c alpha (Lemma 2 then...	Memory beyond the theory window is unnecessary if C2-floor-05 is within CI of C1-where (no floor) for lora and hyper on $\geq$ 5/6 dev systems, on id and sequence_ood....	**FAIL** (same)	no loss vs no floor (HyperKKL, id & seq): 4/6 (need 5/6); TOST(10%) non-inferior: 1/6; probe decays at $\geq$ floor rate: 0/6 systems
C2-window-a1	With a raw input window of L samples instead of the full-history SFM context, how long must the window be?	Supported if e(L $\geq$ L*) is within 5% of e(512) on $\geq$ 5/6 dev systems, on id and sequence_ood.	**FAIL**	e(L $\geq$ L*) within 5% of e(512): id 2/5; seq 3/5 (need 5/6). Refined L* = (ln(1/eta)+ln l)/alpha (post hoc, not pre-registered): id 2/3; seq 2/3
C2-window-a2	Does the window knee move as 1/alpha (Theorem: L* ~ (ln(1/eta) + ln l)/alpha)?	Compare knees with C2-window-a1 on vdp, duffing, battery; supported if the knee in samples changes by $\leq$ 2x on $\geq$ 2/3 systems.	**FAIL**	knee in samples changes by $\leq$ 2x: 1/2 (need 2/3)
C3-faults	Against EKF, UKF, PF, model-based KKL and NLOX (nominal model, oracle noise for clean), and against robust (Huber) and condition-oracle filters,...	Per condition, count dev systems where the best learned observer beats the best filter (nominal, and separately the best of robust/oracle) with a paired CI excluding...	REPORT. (same)	learned-wins regime ($\geq$ 4/6 vs the best filter of either kind, shape and seq): none of 15 conditions
C3-noise-ct	The sensor's noise is colored and heavy-tailed in training and deployment. Filters assume white Gaussian noise (nominal R; oracle filters use its...	Best learned vs best filter of each kind, paired CI on id and sequence_ood, $\geq$ 4/6.	**FAIL** (same)	id: vs best filter 0/6, vs best working 0/6; seq: vs best filter 0/6, vs best working 0/6; failed filters per system 0-1
C3-noise-ct (b)	Does the KKL structure help beyond learning from real-plant data? (best KKL vs best data-driven baseline trained on the same data)	Not pre-registered (added after review E1/E2): best KKL beats the best data-driven baseline (nlox, koopman_kf[_bilinear], greybox_ekf/ukf/ekf_oracle) with a paired...	**FAIL** (same)	best KKL vs best data-driven: id 0/6; seq 0/6 (merged runs 18/18)
C3-mismatch10	Filters use the nominal model; learned observers are trained on data from the real plant. When does learning from data beat a wrong model?	Learned wins at a level if the best learned observer beats the best filter with CI excluding 0 on $\geq$ 4/6 systems, on id and sequence_ood.	**PASS** (same)	id: vs best filter 5/6, vs best working 5/6; seq: vs best filter 4/6, vs best working 4/6; failed filters per system 0-7
C3-mismatch10 (b)	Does the KKL structure help beyond learning from real-plant data? (best KKL vs best data-driven baseline trained on the same data)	Not pre-registered (added after review E1/E2): best KKL beats the best data-driven baseline (nlox, koopman_kf[_bilinear], greybox_ekf/ukf/ekf_oracle) with a paired...	**FAIL** (same)	best KKL vs best data-driven: id 0/6; seq 0/6 (merged runs 18/18)
C3-mismatch20	Filters use the nominal model; learned observers are trained on data from the real plant. When does learning from data beat a wrong model?	Learned wins at a level if the best learned observer beats the best filter with CI excluding 0 on $\geq$ 4/6 systems, on id and sequence_ood.	**PASS** (same)	id: vs best filter 5/6, vs best working 5/6; seq: vs best filter 5/6, vs best working 5/6; failed filters per system 0-8

Table 1: (continued)

node	question	rule (verbatim, abridged)	outcome	evidence (abridged)
C3-mismatch20 (b)	Does the KKL structure help beyond learning from real-plant data? (best KKL vs best data-driven baseline trained on the same data)	Not pre-registered (added after review E1/E2): best KKL beats the best data-driven baseline (nlox, koopman_kf[_bilinear], greybox_ekf/ukf/ekf_oracle) with a paired...	**FAIL** (same)	best KKL vs best data-driven: id 1/6; seq 1/6 (merged runs 18/18)
C3-mismatch30	Filters use the nominal model; learned observers are trained on data from the real plant. When does learning from data beat a wrong model?	Learned wins at a level if the best learned observer beats the best filter with CI excluding 0 on $\geq$ 4/6 systems, on id and sequence_ood.	**PASS** (same)	id: vs best filter 5/6, vs best working 5/6; seq: vs best filter 5/6, vs best working 5/6; failed filters per system 0-8
C3-mismatch30 (b)	Does the KKL structure help beyond learning from real-plant data? (best KKL vs best data-driven baseline trained on the same data)	Not pre-registered (added after review E1/E2): best KKL beats the best data-driven baseline (nlox, koopman_kf[_bilinear], greybox_ekf/ukf/ekf_oracle) with a paired...	**FAIL** (same)	best KKL vs best data-driven: id 0/6; seq 0/6 (merged runs 18/18)
C3-mismatch50	Filters use the nominal model; learned observers are trained on data from the real plant. When does learning from data beat a wrong model?	Learned wins at a level if the best learned observer beats the best filter with CI excluding 0 on $\geq$ 4/6 systems, on id and sequence_ood.	**PASS** (same)	id: vs best filter 6/6, vs best working 6/6; seq: vs best filter 5/6, vs best working 5/6; failed filters per system 0-8
C3-mismatch50 (b)	Does the KKL structure help beyond learning from real-plant data? (best KKL vs best data-driven baseline trained on the same data)	Not pre-registered (added after review E1/E2): best KKL beats the best data-driven baseline (nlox, koopman_kf[_bilinear], greybox_ekf/ukf/ekf_oracle) with a paired...	**FAIL** (same)	best KKL vs best data-driven: id 0/6; seq 0/6 (merged runs 18/18)
C3-mismatch (crossover)	Where does the filter-vs-learned crossover sit?	Hypothesis of C3-mismatch*: crossover between 10% and 30% mismatch (learned wins at a level per that node's rule).	**FAIL** (same)	level outcomes: 10% PASS, 20% PASS, 30% PASS, 50% PASS; first winning level 10%
C3-partial	When the modeller knows only part of the dynamics (a missing or simplified term, not a wrong parameter), do learned observers trained on...	Per system, best learned vs best filter (nominal and robust/oracle), paired CI on id and sequence_ood; report against the relative model error median |f - f_known| / |f|.	REPORT. (same)	id: vs best filter 5/6, vs best working 5/6; seq: vs best filter 5/6, vs best working 5/6; failed filters per system 0-8; hypothesis pattern (learned wins on bioreactor/vdp/lorenz, filters on duffing/quad_tank) holds on 4/5...
C3-partial (b)	Does the KKL structure help beyond learning from real-plant data? (best KKL vs best data-driven baseline trained on the same data)	Not pre-registered (added after review E1/E2): best KKL beats the best data-driven baseline (nlox, koopman_kf[_bilinear], greybox_ekf/ukf/ekf_oracle) with a paired...	**FAIL** (same)	best KKL vs best data-driven: id 0/6; seq 1/6 (merged runs 18/18)
C3-partial-modelfree	Is a partial model in the phase-1 PDE term still worth using, or does it hurt?	Model helps if C3-partial beats this node with CI excluding 0 on $\geq$ 4/6 systems.	**FAIL** (same)	C3-partial (partial model in the PDE term) beats model-free phase 1: id 0/6; seq 0/6 (need 4/6; decided on the compare tier sequence_ood)
C3-realistic	All three kinds of incomplete knowledge at once, as on a real plant.	Best learned vs best filter (any variant), paired CI, id and sequence_ood, $\geq$ 4/6.	**PASS** (same)	id: vs best filter 6/6, vs best working 6/6; seq: vs best filter 6/6, vs best working 6/6; failed filters per system 0-8
C3-realistic (b)	Does the KKL structure help beyond learning from real-plant data? (best KKL vs best data-driven baseline trained on the same data)	Not pre-registered (added after review E1/E2): best KKL beats the best data-driven baseline (nlox, koopman_kf[_bilinear], greybox_ekf/ukf/ekf_oracle) with a paired...	**FAIL** (same)	best KKL vs best data-driven: id 1/6; seq 2/6 (merged runs 30/30)
C3-regimes	Under partial knowledge, do learned KKL observers beat the grey-box EKF/UKF (partial model + learned residual, same data) under a poor initial...	Per condition, best KKL (lora/hyper) vs best grey-box filter (greybox_ekf/ukf/ekf_oracle): paired 95% CI excluding 0 on $\geq$ 4/6 dev systems on sequence_ood = 'KKL-wins...	REPORT. (same)	KKL-wins regimes ($\geq$ 4/6, sequence_ood, vs best grey-box): none of 6 conditions; clean 3/6; poor_init 3/6; noise10 2/6; outliers 1/6; plant_drift10 3/6; burst_outage 1/6

Table 1: (continued)

node	question	rule (verbatim, abridged)	outcome	evidence (abridged)
C3-data50	How much data does a learned observer need before it pays off?	Report error vs n_train (50/150/600) per method; crossover with the filters under 30% mismatch read from C3-mismatch30.	REPORT.	learning curve reported (Figure F3 companion); no pass/fail rule
C3-data150	How much data does a learned observer need before it pays off?	Report error vs n_train (50/150/600) per method; crossover with the filters under 30% mismatch read from C3-mismatch30.	REPORT.	learning curve reported (Figure F3 companion); no pass/fail rule
TEST systems (frozen recipe, run once)				
TEST-C1	Does the dev-system C1 conclusion (weight-space conditioning beats latent injection and the additive 'concat' variant) hold on six plants never...	Supported if the better of lora/hyper beats both latent and concat with a paired CI excluding 0 on $\geq 4/6$ test systems on shape_ood and on sequence_ood. Run once; no...	**PASS** (same)	HyperKKL (r=4) beats latent and concat: shape_ood 5/6; sequence_ood 5/6 (need 4/6 on each)
TEST-C1-tuned	With an equal tuning budget per variant, does weight-space conditioning still beat latent injection and the additive residual?	Tuned HyperKKL (the low-rank hypernetwork, 'lora' code path) beats tuned latent AND tuned concat with a paired 95% CI on $\geq 4/6$ dev and $\geq 4/6$ test systems, shape_ood...	PENDING	node not created yet
TEST-C3	Partial model + 30% parameter error + colored Student-t noise on six held-out plants: does the best learned observer beat every filter variant?	Best learned vs best WORKING filter (NRMSE below climatology), paired CI, id and sequence_ood, $\geq 4/6$; report against the relative model error of each plant. Run once.	**FAIL** orig. PASS	id: vs best filter 3/6, vs best working 3/6; seq: vs best filter 3/6, vs best working 3/6; failed filters per system 0-2
TEST-C3 (b)	Does the KKL structure help beyond learning from real-plant data? (best KKL vs best data-driven baseline trained on the same data)	Not pre-registered (added after review E1/E2): best KKL beats the best data-driven baseline (nlox, koopman_kf[_bilinear], greybox_ekf/ukf/ekf_oracle) with a paired...	**FAIL*** (same)	best KKL vs best data-driven: id 1/4; seq 1/4 (merged runs 14/18)